Omega Axiom

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1 -Axiom-Based Energy Equation and Comparison with Measured Values

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1.1 1. Introduction

The classical relativistic energy-mass relation,

E=mc2,E = m c$\hat{2}$,E=mc2,establishes that mass can be converted into energy.

In this report, we reinterpret the energy relation under the -Axiom framework, where mass depends on a phase function ppp and the effective propagation speed is reduced from the standard light speed CCC to an -universe speed VV_.

We compare the -Axiom-based energy EE_ to conventional relativistic values using electron and proton mass data.

1.2 2. -Axiom Energy Equation

Assuming mass depends on phase ppp and the energy propagation speed is VV_:

E=m(p)V2E_Ω $= m(p) ncdotV\_\Omega\hat{2}E = m(p)V2The-universespeedisdefinedas:$

V=C1+,=0.0412V_Ω $= nfrac\{C\}1 + ndelta\}, nquadndelta = 0.0412V = 1 + C, = 0.0412V2.879108m/sV\_$

E=m(p)C21.0841\boxed{E_Ω $= m(p)ncdotnfrac\{C\hat{2}\}1.0841\}E = m(p)1.0841C2$

1.3 3. Comparison with Measured Values

The following table compares conventional relativistic energy and -Axiom-based energy for electrons and protons:

\begin{table}[h]

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\begin{tabular}lcccc}

\toprule

Particle & Mass mmm (kg) & Relativistic Energy E=mc2E = mc$\hat{2}$E=mc2 (J) & -Axiom Energy EE_ (J) & Ratio E/EE_Ω$/EE/En$

\midrule

Electron & 9.109×10319.109\times10

Proton & 1.673×10271.673\times10

\bottomrule

\end{tabular}

\caption{Comparison of -Axiom-based energy and conventional relativistic energy.}

\end{table}

The -Axiom energy is approximately 7.7–8% lower than the conventional relativistic energy, consistent with the velocity adjustment factor =0.0412\delta = 0.0412=0.0412.

1.4 4. SI vs -Axiom Definitions of Time and Speed

\begin{table}[h]

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\begin{tabular}—l—l—l—}

\hline

\textbf{Category} & \textbf{Standard SI Physics} & \textbf{-Axiom (Revised)} \

\hline

Definition Basis & Cesium-133 atomic 9,192,631,770 oscillations & Photon passing 0.0015 lattice nodes (phase accumulation) \

\hline

Speed Assumption & Light speed CCC is constant (3×1083\times10ˆ83×108 m/s) & Effective light speed depends on lattice density (VV_) \

\hline

Nature of Time & Absolute/relative flow (abstract) & Data latency (physical) \

\hline

1-second Duration & Fixed like a “nail” & Already distorted by factor 1/1.04121/1.04121/1.0412 \

\hline

\end{tabular}

\caption{Comparison of SI standard definitions versus -Axiom-based definitions.}

\end{table}

1.5 5. Conclusion

The -Axiom-based energy equation extends the conventional E=mc2E = mcˆ2E=mc2 relation by considering phase-dependent mass and a reduced effective speed VV_.

The calculated -Axiom energies show an approximate 8% reduction relative to conventional relativistic energies, providing a quantitative check against measured values.

Additionally, the -Axiom framework offers a revised perspective on the definition of time and speed, highlighting physical latency and lattice effects rather than abstract constants.

Future work will incorporate phase-dependent mass corrections m(p)m(p)m(p) and observational or experimental datasets to further validate the -Axiom model.